Preview: set-01/by_level/Grade_03-04_Ecolier/y2024_q19 (v3)

Two repeating patterns together: find the block where both start again, count in it, multiply

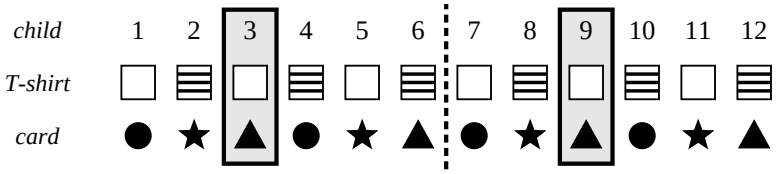
How to think about it: two patterns at the same time

In these questions two things repeat at the same time, for example the pattern on the T-shirt and the shape on the card. You have to find out how often a certain pair happens, like "white T-shirt **and** triangle card".

The big idea: two patterns together make one **longer pattern**. Find the block where **both start again at the same time**, count inside one block, then multiply.

Step by step: the contest question

- **1. Write it down.** Write the numbers 1, 2, 3, ... and draw both patterns under them. Stop when both patterns start again **at the same time**: child 7 looks exactly like child 1.

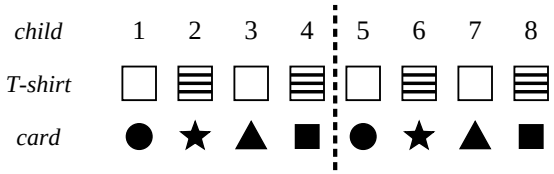


- **2. The block is 6 children long.** After every 6 children everything repeats (the dashed line).
- **3. Count in one block.** In children 1 to 6, only child 3 has a white T-shirt and a triangle card. So there is **1 in every block**.
- **4. Multiply.** 60 children = 60 ÷ 6 = 10 blocks. 10 × 1 = **10** children.

Another way to see it: white T-shirts are children 1, 3, 5, 7, ... (odd numbers). Triangle cards are children 3, 6, 9, 12, ... Both: 3, 9, 15, 21, ... always 6 more.

Watch out

- **The block is not always "one length × the other".** 2 kinds of T-shirt and 4 kinds of card start again together after only **4** children, not 8. Then some pairs **never** happen. Here a striped T-shirt never gets a circle card:



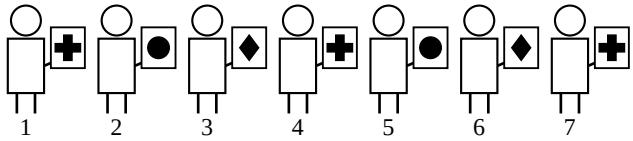
- **Children left over:** 64 children with a block of 6 is 10 blocks (60 children) and 4 more. Check those 4 one by one.

Try it yourself: take a pencil and draw the first 12 children of a pattern. Circle the ones you are looking for. Can you see the block?

Warm-up: small steps first (no points)

W1. Children stand in a row. Their cards repeat like this: cross, circle, diamond, cross, circle, diamond, ...

What shape is on the card of child number 12?



A. cross ***B.*** circle ***C.*** diamond ***D.*** star ***E.*** square

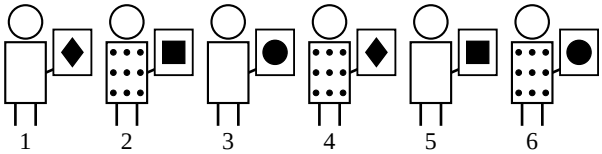
W2. 24 children stand in a row. Their cards repeat like this: triangle, cross, diamond, triangle, cross, diamond, ...

How many have a cross card?

A. 3 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

W3. Here are the first 6 children of a row.

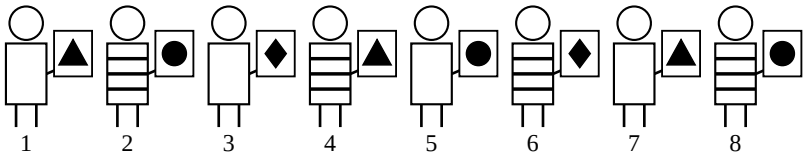
How many of them have a **white** T-shirt **and** a **circle** card?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 6

W4. Look at the children. Child 1 has a white T-shirt and a triangle card.

Which is the **next** child with a white T-shirt and a triangle card?



A. child 7 ***B.*** child 8 ***C.*** child 9 ***D.*** child 10 ***E.*** child 11

W5. In a row of 60 children, the children with a white T-shirt and a star card have the numbers 2, 8, 14, 20, ... (always 6 more).

How many of them are there?

A. 6 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 20

Part A: Each question is worth 3 points

1. 60 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: white, striped, white, striped, ...
- The cards repeat like this: circle, square, diamond, circle, square, diamond, ...

How many children have a **striped** T-shirt **and** a **diamond** card?

A. 5 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 12

2. 70 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: white, striped, white, striped, ...
- The cards repeat like this: circle, cross, diamond, triangle, square, circle, cross, ...

How many children have a **striped** T-shirt **and** a **cross** card?

A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

Part B: Each question is worth 4 points

3. 50 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: black, white, black, white, ...
- The cards repeat like this: triangle, diamond, square, triangle, diamond, square, ...

How many children have a **black** T-shirt **and** a **triangle** card?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 16

4. 73 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: black, striped, dotted, black, striped, dotted, ...
- The cards repeat like this: square, triangle, circle, diamond, star, square, triangle, ...

How many children have a **striped** T-shirt **and** a **star** card?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 14 ***E.*** 24

5. 60 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: white, black, white, black, ...
- The cards repeat like this: circle, square, cross, circle, square, cross, ...

What does child number **54** wear?

A. black T-shirt, cross card ***B.*** white T-shirt, cross card ***C.*** black T-shirt, square card
D. white T-shirt, circle card ***E.*** black T-shirt, circle card

6. 100 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: dotted, black, white, dotted, black, white, ...
- The cards repeat like this: cross, triangle, diamond, star, cross, triangle, ...

What does child number **94** wear?

A. white T-shirt, triangle card ***B.*** dotted T-shirt, triangle card ***C.*** black T-shirt, triangle card
D. dotted T-shirt, diamond card ***E.*** dotted T-shirt, cross card

7. 100 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: striped, dotted, striped, dotted, ...
- The cards repeat like this: square, circle, star, square, circle, star, ...

Count only the children with a **striped** T-shirt and a **circle** card.

Which number in the row does the **5th** of them have?

A. 23 ***B.*** 25 ***C.*** 28 ***D.*** 29 ***E.*** 30

8. 100 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: striped, dotted, striped, dotted, ...
- The cards repeat like this: cross, diamond, square, circle, triangle, cross, diamond, ...

Count only the children with a **dotted** T-shirt and a **triangle** card.

Which number in the row does the **6th** of them have?

A. 60 ***B.*** 61 ***C.*** 65 ***D.*** 70 ***E.*** 72

Part C: Each question is worth 5 points

9. 28 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: white, striped, white, striped, ...
- The cards repeat like this: circle, cross, triangle, star, circle, cross, ...

How many children have a **striped** T-shirt **and** a **circle** card?

A. 0 ***B.*** 3 ***C.*** 6 ***D.*** 7 ***E.*** 14

10. 24 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it.

- The T-shirts repeat like this: black, white, dotted, striped, black, white, ...
- The cards repeat like this: star, triangle, star, triangle, ...

How many children have a **striped** T-shirt **and** a **star** card?

A. 0 ***B.*** 1 ***C.*** 3 ***D.*** 6 ***E.*** 12

11. A big lamp flashes every 2 seconds and a small lamp flashes every 5 seconds. They flash together right at the start.

How many times do they flash **together** in the next 50 seconds? (Count a flash exactly at second 50, but not the one at the start.)

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

12. A big lamp flashes every 4 seconds and a small lamp flashes every 6 seconds. They flash together right at the start.

How many times do they flash **together** in the next 72 seconds? (Count a flash exactly at second 72, but not the one at the start.)

A. 6 ***B.*** 7 ***C.*** 12 ***D.*** 18 ***E.*** 30

Answer Key

W1.	**C**	**W2.**	**E**	**W3.**	**B**	**W4.**	**A**	**W5.**	**D**	**1.**	**D**
2.	**E**	**3.**	**D**	**4.**	**A**	**5.**	**A**	**6.**	**B**	**7.**	**D**
8.	**A**	**9.**	**A**	**10.**	**A**	**11.**	**E**	**12.**	**A**		

Solutions

W1. C (diamond) — The shapes repeat every 3 children. 12 = 3 × 3 + 3, so it is the same as child 3: diamond. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_one_pattern]

W2. E (8) — In every 3 children there is one cross card. 24 ÷ 3 = 8. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_how_many_one]

W3. B (1) — Look at each child: only child number 3. So 1. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_first_block]

W4. A (child 7) — The T-shirt pattern starts again at 3, 5, 7, ... and the card pattern at 4, 7, 10, ... Both together: child 7. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_same_again]

W5. D (10) — One in every block of 6: 60 ÷ 6 = 10. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · w_skip_count]

1. D (10) — Both patterns start again together every 6 children. In each block of 6 only child number 6 (then 12, 18, ...) has a striped T-shirt and a diamond card. 60 ÷ 6 = 10 blocks, so 10. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · both_count]

2. E (7) — Both patterns start again together every 10 children. In each block of 10 only child number 2 (then 12, 22, ...) has a striped T-shirt and a cross card. 70 ÷ 10 = 7 blocks, so 7. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · both_count]

3. D (9) — The patterns start again together every 6 children; in each block child number 1 matches. 50 = 8 × 6 + 2: 8 full blocks give 8, and in the 2 children left over child 49 matches too. 8 + 1 = 9. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · leftover]

4. A (5) — The patterns start again together every 15 children; in each block child number 5 matches. 73 = 4 × 15 + 13: 4 full blocks give 4, and in the 13 children left over child 65 matches too. 4 + 1 = 5. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · leftover]

5. A (black T-shirt, cross card) — The T-shirts repeat every 2: 54 = 26 × 2 + 2, so the T-shirt is the 2nd one: black. The cards repeat every 3: 54 = 17 × 3 + 3, so the card is a cross. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · child_number]

6. B (dotted T-shirt, triangle card) — The T-shirts repeat every 3: 94 = 31 × 3 + 1, so the T-shirt is the 1st one: dotted. The cards repeat every 4: 94 = 23 × 4 + 2, so the card is a triangle. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · child_number]

7. D (29) — The first one is child 5. The patterns start again together every 6 children, so the next ones are 11, 17, ... The 5th: 5 + 4 × 6 = 29. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · kth_match]

8. A (60) — The first one is child 10. The patterns start again together every 10 children, so the next ones are 20, 30, ... The 6th: 10 + 5 × 10 = 60. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · kth_match]

9. A (0) — Write the first children down: both patterns start again together already after 4 children (not 2 × 4 = 8). In each block of 4 no child has a striped T-shirt and a circle card, so the answer is 0. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · never_together]

10. A (0) — Write the first children down: both patterns start again together already after 4 children (not 4 × 2 = 8). In each block of 4 no child has a striped T-shirt and a star card, so the answer is 0. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · never_together]

11. E (5) — The big lamp flashes at 2, 4, 6, ... and the small one at 5, 10, 15, ... The first time both flash is at second 10, then every 10 seconds. 50 ÷ 10 = 5. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · lamps]

12. A (6) — The big lamp flashes at 4, 8, 12, ... and the small one at 6, 12, 18, ... The first time both flash is at second 12 (not 4 × 6 = 24), then every 12 seconds. 72 ÷ 12 = 6. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · lamps]